Advanced Programming Practice · 21CSC203P

Unit 4 · Session 1 · SLO 2 — Solved Worksheet

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Unit â€“ 4 : Session â€“ 1 : SLO â€“ 2 | Functional vs Imperative Paradigms

I. Discussion Question

• A pure functional approach is better for adding two numbers because the operation is a pure function: given the same two inputs it always returns the same result, with no dependence on â€” and no modification of â€” any external state.

• Predictability: add(2, 3) is always 5, regardless of when or how many times it is called, whereas an imperative version may read or mutate global variables, so results can change with program state.

• Testability: pure functions need no setup or mocks â€” they can be unit-tested in isolation.

• Referential transparency: any call can be replaced by its result, enabling safe refactoring, caching (memoization) and parallelism.

• No side effects: the calculator's other features can never be accidentally corrupted by the addition logic, which is a real risk with imperative state mutation.